

Effect of Perturbative Vibronic Correction for Spin–Orbit Coupling in Hot Exciton OLED Systems

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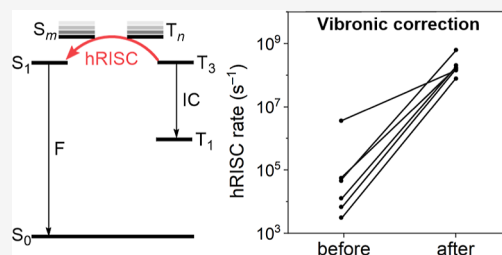


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ABSTRACT: Hot exciton materials have recently attracted attention for their potential to improve the efficiency of fluorescent organic light-emitting diodes (OLEDs) by harvesting triplet excitons through high-lying reverse intersystem crossing (hRISC). As internal conversion (IC) typically proceeds quite fast, an efficient hRISC pathway will be essential for the device performance. Vibronic coupling is one of the key mechanisms known to enhance RISC in conventional donor–acceptor type thermally activated delayed fluorescence (TADF) molecules. Motivated by this, we evaluate the vibronic correction of spin–orbit coupling (SOC) in 10′-diphenyl-9,9′-bianthracene (PPBA) and its derivatives using a first-order perturbative approach based on time-dependent density functional theory (TDDFT). Although the target states T_3 and S_1 are close in energy, their similar charge-transfer (CT) character results in negligibly small zeroth-order SOC, leading to intrinsically slow hRISC rates in all considered molecules. However, vibronic corrections significantly enhance SOC and increase the hRISC rate by nearly 4 orders of magnitude. At vibrationally distorted geometries, we observe partial mixing of CT and locally excited (LE) character in a nonidentical fashion between the two electronic states, thereby dynamically enhancing SOC. These findings demonstrate the importance of vibronic effects in hot exciton systems with flexible multiple-ring backbones and highlight the useful correction schemes in photophysical analyses of OLED materials.



1. INTRODUCTION

Improving the efficiencies of organic light-emitting diode (OLED) systems is an ongoing challenge. Since excitons in OLED are formed electronically, the ratio between singlet and triplet excitons is 1:3 by spin statistics. Because only singlet excitons are emissive, triplet excitons should somehow be utilized to boost the device efficiency. Phosphorescent emitters, often containing expensive heavy metal elements, are commonly adopted for this purpose. Fluorescent emitters are also employed to avoid using such heavy metals, with the help of some triplet harvesting processes such as thermally activated delayed fluorescence (TADF), triplet–triplet fusion (TTF), and hot exciton processes.

Among the three processes, the hot exciton one involves harvesting through reverse intersystem crossing (RISC), originating not from T_1 but from higher-lying triplet states (hence referred to as “hRISC”). Although the hot exciton process has nonconventional and somewhat unusual anti-Kasha characteristics, numerous hot exciton materials have been reported with several noticeable advantages.^{1–10} For example, unlike TADF, the character of the S_1 state in a hot exciton molecule does not rely on charge transfer (CT). Instead, it has local excitation (LE) or hybridized local and charge transfer (HLCT) characters. Therefore, hot exciton emitters can have high photoluminescence quantum yield (PLQY) with fast fluorescence, while conventional TADF emitters tend to have relatively low PLQY due to diminished

transition dipole size inherited from the dominant CT character. In addition, the populations of long-lived triplet excitons are significantly reduced due to fast hRISC. In fact, long-lived triplet excitons can deteriorate efficiency by inducing exciton decays such as singlet–triplet annihilation and triplet–polaron quenching. Moreover, long-lived triplet excitons can even increase the chances of chemical bond breaking. Therefore, suppressing the population growth of triplet excitons can improve both stability and efficiency.

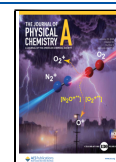
The most important issue in the hot exciton process is the competition between hRISC and internal conversion (IC) to T_1 . In principle, a hRISC process may commence from any triplet states, but in many cases, it appears to initiate from T_2 and migrate into S_1 .^{11–14} Therefore, the basic strategy for designing hot exciton materials is to increase the energy gap between T_2 and T_1 . However, the T_2 – T_1 energy gap tends to be at most 2 eV and is usually smaller than the visible gap. This poses a great challenge, as the radiative decay lifetimes of fluorescent emitters are typically in the nanosecond regime,

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while the IC time scales for such small gaps are likely be faster than a nanosecond. As a matter of fact, when we calculated hRISC time scales for a series of hot exciton molecules with a time-correlation and harmonic normal-mode based approach,¹⁵ the results were always slower than a nanosecond. To the best of our knowledge, there has been no computational evidence explicitly showing that the hRISC rate is faster than, or at least comparable to, the IC rate.^{16–19}

Nevertheless, experimental evidence indicates that materials exhibiting the hot exciton properties possess unique characteristics that are not easily explained by TADF or TTF mechanisms. Perhaps, the discrepancy may have arisen from the limitation of harmonic oscillator models, which can become unreliable with substantial conformational relaxations associated with electronic transitions. There is also a possibility that additional mechanisms, such as transition through multiple hRISC channels, are involved.^{20,21} In addition, the reversible charge recombination processes in the early stage of exciton formation have been overlooked.²² Namely, if charge recombination rates into excitons depend on spin multiplicities, and if the spin statistics influences only up to the charge pair formation, the conventional 1:3 ratio rule may be contravened,^{23,24} as long as the spin crossover in the charge pair state is fast enough. In addition, higher order perturbations that enhance hRISC have not yet been considered in an appropriate manner. In fact, many hot exciton molecules, if not all, tend to have flexible linker moieties, and second order rate improvements^{25–27} induced by molecular vibrations may turn out to be important.

In this study, we focus on vibration-related second order effects on hRISC rates in hot exciton molecules. Recently, some of the authors have reported that a similar vibronic effect can notably enhance fluorescence rates in donor–acceptor TADF systems by effectively increasing the transition dipole moment.²⁸ The method was based on the first-order perturbative wave function correction, and can easily be extended to calculate spin–orbit coupling (SOC). Indeed, the related spin-vibronic effect has been reported as essential in phosphorescent or TADF emitters.^{29–31} Here, we demonstrate that spin-vibronic effects on hRISC rates are indeed dramatic by considering 10'-diphenyl-9,9'-bianthracene (PPBA)³² and its derivatives (see Figure 1 for the molecular structures). Utilizing the thermal vibration correlation function (TVCF) formalism,^{15,31,33} we compute both hRISC rates with and

without vibronic coupling, and also IC rates for comparative analyses. We discuss the factors affecting such large vibronic coupling and conclude with future aspects that could be pursued for further developing hot exciton molecules.

2. THEORY

2.1. Perturbative Expansion for Spin–Orbit Coupling and hRISC Rate

The theoretical framework used in this study is based on our previous formulation of perturbative vibronic corrections, which was developed to describe the transition dipole moments and radiative rates of TADF emitters.²⁸ Here, we extend it to SOC and employ it to quantify vibronically corrected hRISC processes in hot-exciton systems.

A brief exposition of the first-order perturbative wave function correction is presented here, while more detailed information can be found in earlier works, particularly in ref 28. We start with the first-order perturbative wave function correction

$$|\Psi_n^{(1)}\rangle = \sum_{m \neq n} \frac{\langle \Psi_m^{(0)} | \hat{H}' | \Psi_n^{(0)} \rangle}{E_n^{(0)} - E_m^{(0)}} |\Psi_m^{(0)}\rangle \quad (1)$$

where m and n denote the indices of electronic states. When the zeroth-order Hamiltonian H_0 is defined at the equilibrium geometry and the perturbation Hamiltonian H' is induced by a geometric distortion, the total Hamiltonian can be written as

$$\hat{H}(r, R) \approx \hat{H}(r, R_0) + (R - R_0) \frac{\partial \hat{H}}{\partial R} \quad (2)$$

Substituting the perturbation into eq 1, the total wave function will expand as

$$|\Psi_n\rangle \approx |\Psi_n^{(0)}\rangle + \sum_{m \neq n} |\Psi_m^{(0)}\rangle \left\langle \Psi_m^{(0)} \left| \frac{\partial}{\partial R} \right| \Psi_n^{(0)} \right\rangle \Delta R \quad (3)$$

With this, SOC between S_m and T_n is written to the first order as

$$\begin{aligned} \langle S_m | \hat{H}_{\text{SO}} | T_{n,m_s} \rangle &\approx \langle S_m^{(0)} | \hat{H}_{\text{SO}} | T_{n,m_s}^{(0)} \rangle \\ &+ \sum_{i \neq n} \langle S_m^{(0)} | \hat{H}_{\text{SO}} | T_{i,m_s}^{(0)} \rangle \left\langle T_i^{(0)} \left| \frac{\partial}{\partial R} \right| T_n^{(0)} \right\rangle \Delta R \\ &- \sum_{i \neq m} \left\langle S_m^{(0)} \left| \frac{\partial}{\partial R} \right| S_i^{(0)} \right\rangle \langle S_i^{(0)} | \hat{H}_{\text{SO}} | T_{n,m_s}^{(0)} \rangle \Delta R \\ (m_s = 0, \pm 1) \end{aligned} \quad (4)$$

The number of intermediate excited states indexed by i should be reasonably large for reliable vibronic corrections. A numerical test with PPBA found that the same value of $i_{\text{max}} = 10$ as adopted in our previous study²⁸ was again reasonable, and we adopted this limit throughout this study (see Section S1 for this numerical test.)

The geometric displacement, ΔR , was calculated as the thermal motion of the molecule. We adopted the normal mode picture to calculate the thermally averaged motion. The vibrational energy is written as

$$E_v = G \cdot \Delta R + \frac{1}{2} \Delta R \cdot H \cdot \Delta R \quad (5)$$

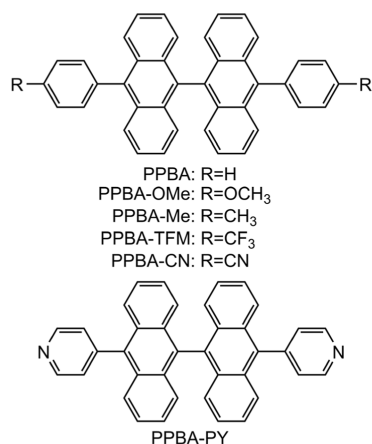


Figure 1. Chemical structures of PPBA and its derivatives.

where G is the gradient and H is the Hessian. If we implement our approach only around the equilibrium geometry, G becomes zero by construction. We can rewrite eq 5 with the mass-weighted Hessian and coordinates, defined as $F = M^{-1/2}HM^{-1/2}$ and $\delta q = M^{1/2}\Delta R$ with the diagonal mass matrix M

$$E_v = \frac{1}{2}\delta q \cdot F \cdot \delta q = \frac{1}{2}(\delta q U) \text{diag}(\omega_1^2, \omega_2^2, \dots, \omega_N^2)(U^\dagger \delta q) \quad (6)$$

with the unitary matrix U associated with the normal mode vectors, $U = (|v_1\rangle, |v_2\rangle, \dots, |v_N\rangle)$. If δq is a constant multiple of the j th normal mode vector given as $\delta q = a_j |v_j\rangle$, the right-hand side of eq 6 will be the sum of $\bar{E}_j \equiv a_j^2 \omega_j^2 / 2$, where a_j physically represents the scalar vibrational displacement. To obtain the geometric displacement in a statistically meaningful way, we can consider the Boltzmann distribution $P(a_j) \sim \exp(-a_j^2 \omega_j^2 / 2k_B T)$ for a given frequency ω_j at a temperature of $T = 300$ K. Mathematically, sampling the vibrational displacement is equivalent to generating Gaussian random numbers with the standard deviation $\sigma_j = \sqrt{k_B T / \omega_j^2}$. As normal mode motions are independent of each other, the k th random geometric displacement is generated as

$$\Delta R_k = \sum_j^{3N-6} a_{j,k} M^{-1/2} |v_j\rangle \quad (7)$$

Substituting this displacement into eq 4, we can yield the SOC in the k th randomly distorted geometry. The final SOC value is calculated as a thermal average

$$V_{\text{SO}}^{\text{cor}} = \frac{1}{k_{\text{max}}} \sum_k^k \sqrt{|(S_m | \hat{H}_{\text{SO}}^k | T_{n,0})|^2 + |(S_m | \hat{H}_{\text{SO}}^k | T_{n,1})|^2 + |(S_m | \hat{H}_{\text{SO}}^k | T_{n,-1})|^2} \quad (8)$$

Practically, $k_{\text{max}} = 10^6$ was a sufficiently large number to reach a convergence.

The calculation of nonradiative transition rates follows the theoretical framework detailed in ref 33, and is based on Fermi's golden rule and the correlation function formalism.^{15,31,33} The rate constant k_{hRISC} and the lifetime τ_{hRISC} for hRISC processes from T_n to S_m are given by

$$k_{\text{hRISC}} = \frac{1}{\tau_{\text{hRISC}}} = \frac{1}{\hbar Z} |V_{\text{SO}}|^2 \int_{-\infty}^{\infty} \exp(-i\Delta E_{mn}\tau) \rho(\tau, \tau') d\tau \quad (9)$$

where Z is the vibrational partition function and $\rho(\tau, \tau')$ is the correlation function with $\tau \equiv t/\hbar$ and $\tau' = -\tau - i/k_B T$. This correlation function is expressed as

$$\rho(\tau, \tau') = \text{Tr}[\exp(-i\hat{H}_m\tau)\exp(-i\hat{H}_n\tau')] \quad (10)$$

with $\hat{H}_n$ and $\hat{H}_m$ respectively representing the $(3N - 6)$ -dimensional harmonic oscillator Hamiltonians for the initial and the final states, where N is the number of atoms in the given molecule.

Similarly, the rate constants k_{IC} and lifetimes τ_{IC} for IC processes from T_n to T_m are calculated to allow a direct comparison with the RISC process

$$k_{\text{IC}} = \frac{1}{\tau_{\text{IC}}} = \frac{1}{\hbar Z} \int_{-\infty}^{\infty} \exp(-i\Delta E_{mn}\tau) \sum_{\kappa, \lambda} T_{\kappa, \lambda} \rho_{\text{IC}}(\tau, \tau'; \kappa, \lambda) d\tau \quad (11)$$

The matrix elements $T_{\kappa, \lambda}$ are defined as

$$T_{\kappa, \lambda} = \langle \phi_m | \hat{P}_\kappa | \phi_n \rangle \langle \phi_n | \hat{P}_\lambda | \phi_m \rangle \quad (12)$$

where $\hat{P}_\kappa \equiv -i\hbar\partial/\partial Q_{m, \kappa}$ and $\hat{P}_\lambda \equiv -i\hbar\partial/\partial Q_{n, \lambda}$ are the momentum operators of the κ th normal mode of state T_m and the λ th normal mode of state T_n respectively. $|\phi_n\rangle$ and $|\phi_m\rangle$ are the electronic wave functions of the initial and the final states, respectively. Finally, the correlation function for IC, $\rho_{\text{IC}}(\tau, \tau'; \kappa, \lambda)$, is given by

$$\rho_{\text{IC}}(\tau, \tau'; \kappa, \lambda) = \text{Tr}[\hat{P}_\kappa \exp(-i\hat{H}_n\tau') \hat{P}_\lambda \exp(-i\hat{H}_m\tau)] \quad (13)$$

2.2. Computational Details

We adopted the parameter-tuned ω B97X-D functional³⁴ with the 6-31G(d) basis set for all time-dependent density functional theory (TDDFT) calculations within the Tamm-Dancoff approximation (TDA) (see Section S2 for details of the functional choice).^{35–37} The range-separation parameter ω was tuned to minimize J^2 , defined as

$$J^2 = \sum_{i=0}^1 [\varepsilon_{\text{H}}(N+i) + \text{IP}(N+i)]^2 \quad (14)$$

where ε_{H} is the HOMO energy, IP is the ionization potential, and N is the number of electrons in the given molecule.^{38–41} Because ω depends on the molecular geometry, both ω and the geometry were optimized self-consistently in the ground state (see Section S3 for the optimized parameters). In addition, ω was optimized under the polarized continuum model (PCM) condition^{42,43} with the dielectric constant of 4,4'-N,N'-dicarbazolebiphenyl (CBP, $\varepsilon = 3.5$), which was adopted as the host material.³² All computations were conducted with Q-Chem 6.0.⁴⁴ The optimized ground-state (S_0) geometries were confirmed to be energy minima through the harmonic frequency analysis. However, some excited-state optimizations resulted in normal modes with very small imaginary frequencies (see Section S4), likely due to geometrical flexibilities together with the numerical error associated with diagonalization. To circumvent this issue, we replaced such imaginary frequencies with half the value of the smallest real frequency in the system. Indeed, as described in Section S5A with Table S6, the corrected SOC values displayed very low sensitivity to this frequency as long as it was low enough. Further discussion regarding this treatment can be found in the same section.

3. RESULTS AND DISCUSSION

3.1. Transition Rates of hRISC with Uncorrected and Corrected Spin–Orbit Coupling

Before computing SOC, it was essential to identify which transitions would be suitable for hRISC. For fast hRISC, the energy gap between the selected T_n and its matching singlet state at the T_n -optimized geometry should be small. Additionally, higher-lying states are more likely to suffer from competition with IC. Thus, it will be reasonable to choose the lowest possible n -value, and we selected T_3 and S_1 as the

initial and the final states, with their narrow energy gap of ~ 0.006 eV (see Section S6 for consideration of other hRISC channels). The energy levels of the ten lowest singlet and triplet states at the T_3 -optimized geometry are shown in Figure 2. Other PPBA derivatives exhibited similar energy level

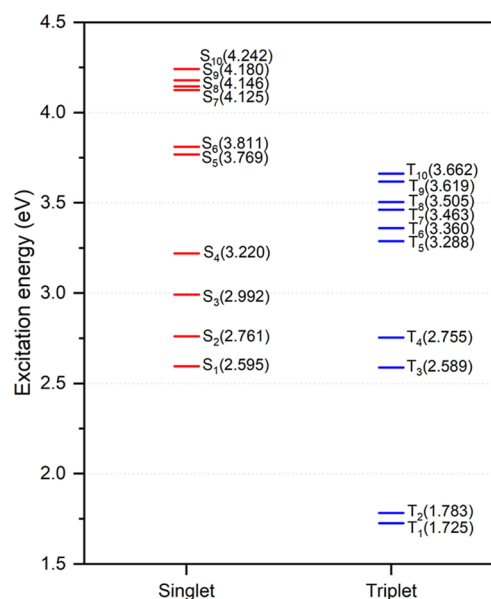


Figure 2. Excitation energies of the ten lowest singlet and triplet states for PPBA at the T_3 -optimized geometry.

structures (see Section S7) and we assumed that the hRISC processes in these derivatives would also initiate from T_3 . Both S_1 and T_3 display significant CT characters, as indicated by their natural transition orbitals (NTOs) shown in Figure S6, with charge transfer from bottom anthracene moiety to the other one. Additionally, in all cases, the electron donor and the acceptor are oriented perpendicular to each other.

Because both T_3 and S_1 states have very similar electronic structures with nearly identical transition orbitals, the SOC value between the two states is small, as dictated by the El-Sayed rule.⁴⁵ The uncorrected SOC of PPBA is 0.001 cm^{-1} as shown in Table 1, indicating that hRISC from T_3 to S_1 would be negligibly slow. For other PPBA derivatives, the uncorrected SOC values are also quite small, ranging from 0.001 cm^{-1} (PPBA-Me and PPBA-TFM) to 0.025 cm^{-1} (PPBA-OMe). In typical TADF molecules, SOC values are at least 0.1 cm^{-1} . Because IC within the triplet manifold of PPBA occurs quite rapidly, even more rapidly than the nonradiative decays from S_1 states of typical TADF systems, SOC values even up to around 0.1 cm^{-1} would still be insufficient to enable efficient hRISC. Indeed, the

calculated IC rates from T_3 to T_1 were 4–7 orders of magnitude larger than the hRISC rates based on these SOC values. Although the calculated rates should surely bear some uncertainties, this large discrepancy between hRISC and IC rates would suggest that hRISC would be unlikely.

On the other hand, the corrected SOC of PPBA is more than 170 times larger than the uncorrected value. Because the transition rate is proportional to the square of SOC, the predicted hRISC rate after the correction is more than 4 orders of magnitude faster than the uncorrected one. The other PPBA derivatives display similar behaviors, with their SOC values increasing from 6 times (PPBA-OMe) to 146 times (PPBA-TFM). Especially, PPBA-CN exhibits the largest corrected SOC value (0.270 cm^{-1}), resulting in the fastest corrected hRISC rate. This enhancement arises from an intermediate state that simultaneously provides large SOC and derivative coupling (see Figure S9e). Overall, the corrected hRISC rates of PPBA derivatives will be 2–5 orders of magnitude faster than the uncorrected rates.

The calculated IC lifetimes for PPBA and its derivatives are consistently found within the range of 10^{-10} – 10^{-11} s, indicating fast relaxation within the triplet manifold. Although the corrected hRISC rates are still 1–2 orders of magnitude slower than the IC rates, the gap between the two processes is significantly reduced compared to that before the correction. Notably, PPBA-Me and PPBA-PY, which exhibit relatively longer IC lifetimes (on the order of 10^{-10} s), exhibit a 15-fold difference between hRISC and IC. These findings indicate that perturbative vibronic corrections are indeed essential for handling hRISC processes at least for PPBA and its derivatives.

3.2. Factors Affecting the Extent of Vibronic Corrections on Spin–Orbit Coupling

Up to this point, through the help of perturbative correction as in eq 4, we have seen that vibrational distortions (ΔR) in PPBA and PPBA-derivatives enhance SOC to large extents. In fact, directly calculating SOC values at different geometries with varying ΔR values will also be possible, but with as many as $k_{\text{max}} = 10^6$ geometries, doing so will be computationally too demanding. In contrast, the perturbative correction scheme essentially requires only a small number of quantum chemical calculations, and the associated computational economy was the main reason we adopted the scheme. Even still, it will be prudent to perform direct SOC calculations and compare them with the perturbation-based results, at least for a manageable small number of samples. Thus, we explicitly computed excitation energies as well as SOC values between S_1 and T_3 states using TDDFT at vibrationally distorted geometries of PPBA, by adopting 100 randomly selected geometries out of the 10^6 structures stated in the previous section.

Table 1. Calculated Photophysical Properties of PPBA and Its Derivatives

compound	spin–orbit coupling		hRISC lifetimes		
	$V_{\text{SO}}^{(0)} (\text{cm}^{-1})^a$	$V_{\text{SO}}^{\text{cor}} (\text{cm}^{-1})^b$	$\tau_{\text{hRISC}}^{(0)} (\text{s})^a$	$\tau_{\text{hRISC}}^{\text{cor}} (\text{s})^b$	$\tau_{\text{IC}}^{(0)} (\text{s})^c$
PPBA	0.001	0.170	1.30×10^{-4}	6.40×10^{-9}	6.22×10^{-11}
PPBA-OMe	0.025	0.149	2.47×10^{-7}	7.18×10^{-9}	4.38×10^{-11}
PPBA-Me	0.001	0.139	2.77×10^{-4}	1.29×10^{-8}	8.07×10^{-10}
PPBA-TFM	0.001	0.146	6.69×10^{-5}	5.01×10^{-9}	4.41×10^{-11}
PPBA-PY	0.003	0.138	1.53×10^{-5}	5.39×10^{-9}	3.53×10^{-10}
PPBA-CN	0.003	0.270	1.64×10^{-5}	1.57×10^{-9}	3.34×10^{-11}

^aZeroth-order term from T_3 to S_1 . ^bCorrected with $i_{\text{max}} = 10$ electronic states. ^cInternal conversion lifetimes from T_3 to T_1 .

Across the 100 sampled geometries, the perturbatively corrected SOC values were enhanced by factors ranging from 25 to 464 relative to the original value (0.001 cm^{-1} ; see Table S10) calculated at the T_3 -optimized geometry, underscoring the substantial impact of vibrational distortions on spin–orbit coupling. The average corrected SOC of the 100 samples was 0.162 cm^{-1} , in good agreement with the mean over all 10^6 geometries (0.169 cm^{-1}). One caveat in performing explicit TDDFT calculations is the possibility of state switching arising from structural distortions that may alter the ordering in the excited states. To ensure meaningful comparison, we tracked the CT character of the original S_1 and T_3 states with zero displacement using NTO analysis and employed the pair of character-consistent states to benchmark the perturbative scheme (see Figure S7 in Section S9 Supporting Information). The average SOC of the resulting character-tracked TDDFT was 0.125 cm^{-1} —slightly lower than the perturbative estimate of 0.162 cm^{-1} . Although the perturbative method modestly overestimates SOC, the deviation remains acceptable given the significant computational savings. In particular, since k_{hrisc} is proportional to the square of the SOC value (eq 9), this overestimation alters the rate constant by much less than an order of magnitude when we assume the same S_1 – T_3 energy gap. This level of agreement highlights the practical utility of the perturbative correction scheme, especially for applications involving statistical ensemble averaging or large-scale rate estimations where direct TDDFT calculations require substantial computational resources.

Beyond confirming the validity of our perturbative approach, these direct calculations may also provide comprehensive insight into the effect of vibrational perturbations on SOC. As described in Figure 3a, the sample exhibiting the most enhanced SOC shows a corrected value of 0.464 cm^{-1} , while maintaining a small S_1 – T_3 energy gap of -0.021 eV with a reversal in state ordering compared to the T_3 -optimized geometry (0.006 eV). To rationalize this enhancement, we performed additional NTO analysis at this distorted geometry and provided the result in Figure 3b. At the T_3 -optimized geometry, both the S_1 and the T_3 states exhibit CT transitions from the lower to the upper anthracene unit, consisting almost exclusively of a HOMO $\rightarrow$ LUMO transition (99.6%). In contrast, at the sampled distorted geometry, the NTOs of the S_1 state exhibit increased spatial overlap between hole and electron NTOs that span over the two anthracene units. This is due to partial mixing with LE character and contributions beyond a simple HOMO $\rightarrow$ LUMO transition (see Figure S8 for more details). Meanwhile, the T_3 state retains the CT character observed at the T_3 -optimized structure. These unequal changes in the electronic structures of the S_1 and the T_3 states, induced by the vibrational distortion, lead to a greater disparity in their transition characters. As a result, the El-Sayed rule allows spin crossover with much enhanced SOC.

This result suggests that vibrational motion promotes different degrees of orbital mixing depending on spin states, and this differentiation overcomes the inherently small SOC between two pure CT states and thereby facilitates faster hrISC from T_3 to S_1 . It also highlights the critical role of vibronic effects in enhancing SOC—a mechanism often invoked to explain improvements of other OLED materials.^{30,31,46–48}

According to eq 4, the vibronic correction depends on several factors: the size of the zeroth-order SOC between the target state and the intermediate states, the magnitude of the

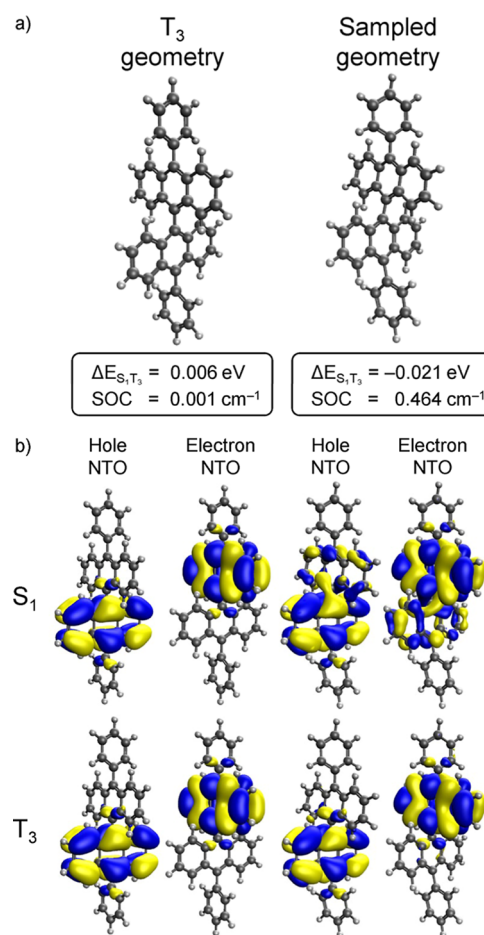


Figure 3. Comparison of T_3 -optimized and thermally sampled geometries of PPBA. (a) Energy gap and spin–orbit coupling between S_1 and T_3 . (b) Natural transition orbital analysis.

derivative coupling vector (DCV), and the tendency of the molecule to move in the direction of the derivative coupling with thermally accessible ΔR . Thus, if we decompose the molecular motions into normal modes, the vibronic correction will be significant when a specific normal mode vector aligns with a sufficiently large DCV especially when the mode frequency is low such that its thermal amplitude is large. The sizes of the zeroth-order SOC, derivative couplings, and vibronic corrections for each intermediate state are shown in Figure 4. In general, as expected, the vibronic correction is large when both SOC and derivative coupling are large. However, the contributions of these two factors are not directly proportional to the overall vibronic corrections because the normal mode contributions behave in a somewhat complex manner. Specifically, when the derivative coupling overlaps primarily with a high-frequency vibration, which is not highly thermally active, the magnitude of DCV does not significantly contribute to the vibronic correction. Nevertheless, SOC and derivative coupling still serve as useful, though not quantitative, indicators in the pursuit of large vibronic corrections.

In this regard, vibrational analyses may provide additional insights into the origin of the vibronic corrections. Figure 5 presents the vibrational mode vectors that significantly influence vibronic perturbations in PPBA and its derivatives. A key vibrational mode in PPBA is a low-frequency mode at 5.8 cm^{-1} . Considering that the thermal energy at room

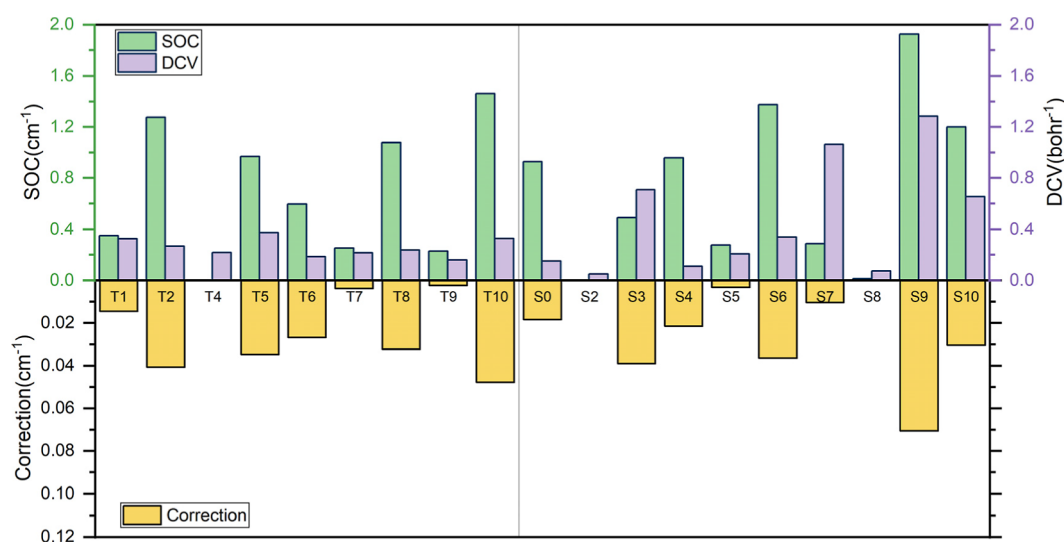


Figure 4. Intermediate-state-resolved analysis of the first-order vibronic correction to the T_3 – S_1 spin–orbit coupling. The triplet-intermediate state contribution (top left panel) shows the zeroth-order spin–orbit coupling (SOC) between S_1 and T_i and the magnitude of derivative coupling vector (DCV) between T_3 and T_i . The singlet-intermediate state contribution (top right panel) shows SOC between S_i and T_3 and DCV between S_1 and S_i . The yellow bars (bottom panel) display the first-order correction by each intermediate state. DCV magnitudes are given as the norms of the derivative coupling vectors. Data for other PPBA derivatives are found in Section S10.

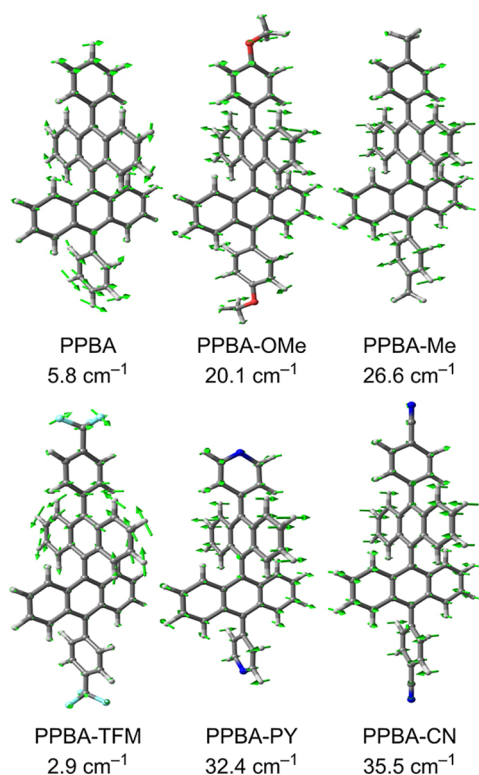


Figure 5. Most important vibrational modes in the first-order correction for PPBA and its derivatives.

temperature is ~ 200 cm^{-1} and that the thermal amplitude of a harmonic oscillator is inversely proportional to its vibrational frequency, such low-frequency modes can induce large geometric displacements (ΔR), which in turn can enhance SOC and thereby accelerate the hRISC process. Notably, the dominant vibrational mode in PPBA and PPBA-TFM involves a rocking motion between the two anthracene moieties, whereas PPBA-OMe, -Me, -PY, and -CN derivatives exhibit twisting motions between the two anthracene moieties. At its

equilibrium geometry, PPBA adopts D_{2d} symmetry, under which the hRISC transition from T_3 to S_1 is symmetry-forbidden. However, vibrational modes that break this geometric symmetry of the PPBA core can induce CT–LE state mixing, thereby enhancing the SOC value.⁴⁹ In the end, the combination of low-frequency modes, geometric displacements, and derivative couplings results in notable vibronic perturbations to overcome negligible zeroth-order SOC caused by identical electronic characters of the two interconvertible states.

3.3. Beyond the First-Order Perturbation Approach on Spin–Orbit Coupling

Here, we discuss the limitations of the first-order perturbative treatment that we have taken toward estimating vibronic corrections to SOC. Although there were significant increases in the hRISC rates of PPBA, the resulting rates still remained at an insufficient level for meaningfully enhancing luminescence efficiency.

A limitation originates from the electronic-structure method itself. Because our mechanistic interpretation depends on the energetic alignment and CT–LE character of several high-lying excited states, the intrinsic limitation of TDDFT for such states introduce unavoidable uncertainties. Particularly, its functional-dependence in treating charge-transfer and near-degenerate states tend to affect the results and care must be taken in analyzing (see Section S2).

Likewise, the implicit PCM model cannot account for specific dopant–host interactions that may perturb excited-state ordering in the solid state. Environmental factors in the solid state such as intermolecular packing and host–guest interactions may also influence hRISC dynamics. Indeed, experimental data have shown high exciton utilization efficiency values ($>25\%$) when PPBA was doped in a CBP host,³² suggesting that interactions with the host or surrounding molecules further promotes hRISC beyond what can be captured in calculations with an isolated molecule.

In addition, our previous study demonstrated that inaccuracies may arise from factors such as anharmonicity

and potential state-switching issues.²⁸ In the case of PPBA and its derivatives shown in the above, the zeroth-order SOC's were negligibly small and could be ignored when calculating the total SOC's. As a result, the total SOC was primarily determined by the first-order terms, which are proportional to the size of the geometric displacement ΔR (eq 4). Under the harmonic approximation, thermal sampling will tend to underestimate ΔR on the large displacement side (see Section SSC). This is because the penalty by a vibrational energy increase is larger with a harmonic potential than with its matching anharmonic potential on that side. Thus, properly including anharmonic effects will likely increase ΔR and thus further enhance the first-order SOC. Because the hRISC rate scales quadratically with SOC, enhancements in the rate by this anharmonicity may become significant. Of course, in cases where thermal vibrational displacements are excessive with increased anharmonicity, the validity of the perturbative approach will diminish and the state switching issue will become immense. In such extreme cases, a completely different computational tactic will need to be designed.

Another aspect that we also need to consider is the possibility of state-switching with large enough ΔR . Actually, our sample analysis with explicit TDDFT calculations exhibited state-switching behaviors in 17 out of the 100 geometries (see Section S9). In these cases, the original CT-type state was elevated as a higher-lying excited state. In some of the distorted geometries, the energy level of the target state pair became closer to another excited state than to each other, suggesting that other states may become involved in the hRISC process under significant geometric distortions. In this respect, the system may not be entirely free from state-mixing issues, particularly near regions of energetic degeneracy. Such possibilities complicate the interpretation of vibrational effects and cannot be reliably inferred from geometry alone. Although the deviation caused by state-switching was limited in our case (recall the difference of ~ 0.037 cm⁻¹ given in Section 3.2), it still highlights the need for caution when applying perturbative treatments and also underscores the importance of validating the approximation.

Despite these limitations, the magnitudes of SOC corrections that we have observed in PPBA and its derivatives are substantial. This suggests that even a first-order perturbative treatment can capture essential features of vibronic coupling in flexible OLED materials. In such systems, thermal vibrational motions, particularly the low-frequency ones, can change electronic structures of excited states by mixing CT and LE characters. In this context, vibronic coupling emerges as a key physical factor in modulating SOC and must be carefully considered in the interpretation and design of hot exciton mechanisms.

4. CONCLUSIONS

Utilizing first-order perturbation theory, we conducted a computational investigation into the vibronic effects on SOC's and hRISC rates with PPBA. We computed derivative couplings and SOC's between the target states (T_3 and S_1) and intermediate excited states at the TDDFT level, while also considering vibrational displacements in the T_3 state under the room-temperature condition. Since both T_3 and S_1 exhibit an identical CT character, the zeroth-order SOC between them was negligibly small, and the hRISC rate appeared to be much slower than that of conventional IC. The vibronic correction enhanced the hRISC rate by approximately up to 4 orders of

magnitude, demonstrating that vibronic effect should not be neglected in flexibly linked systems such as PPBA. Indeed, PPBA derivatives also exhibited significant vibronic corrections.

Our approach involves several approximations that may introduce quantitative errors in the predicted hRISC rates. For instance, we adopted the harmonic approximation to estimate thermal geometric displacement ΔR , and did not fully incorporate possible state-switching behaviors in molecules containing multiple aromatic rings. Nevertheless, we showed that the correction scheme could offer semiquantitative predictions with valuable insights into the nature of hot exciton systems, particularly for flexibly linked emitters. We believe more extended theoretical accounts in the future will further disclose the fundamental nature of hot exciton systems.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jpca.5c05378>.

Optimized geometries and vibrational frequencies of PPBA and its derivatives; details on treating small imaginary frequencies; excitation energies at T_3 -optimized geometries; natural transition orbitals; results from direct TDDFT calculations with 100 randomly sampled geometries; state-specific analyses on the sizes of perturbative corrections. TOC placeholder (PDF)

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Notes

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